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Ex 11: Map Coloring (CSP with Backtracking)

CODE:

```
def is_valid(node, color, assignment, graph):
    for neighbor in graph[node]:
        if neighbor in assignment and assignment[neighbor] == color:
            return False
    return True
def map_coloring(nodes, colors, graph, assignment={}):
    if len(assignment) == len(nodes):
        return assignment
    unassigned = [n for n in nodes if n not in assignment][0]
    for color in colors:
        if is_valid(unassigned, color, assignment, graph):
            assignment[unassigned] = color
            result = map_coloring(nodes, colors, graph, assignment)
            if result:
                return result
            assignment.pop(unassigned)
    return None
nodes = ['WA', 'NT', 'SA', 'Q', 'NSW', 'V', 'T']
colors = ['Red', 'Green', 'Blue']
graph = {
```

```
    'WA': ['NT', 'SA'],
    'NT': ['WA', 'SA', 'Q'],
    'SA': ['WA', 'NT', 'Q', 'NSW', 'V'],
    'Q': ['NT', 'SA', 'NSW'],
    'NSW': ['Q', 'SA', 'V'],
    'V': ['SA', 'NSW'],
    'T': []
}
print(map_coloring(nodes, colors, graph))
```

OUTPUT:

```
============== RESTART: C:/Users/Saiko/Documents/CSA1738/EXP-11.py =============
{'WA': 'Red', 'NT': 'Green', 'SA': 'Blue', 'Q': 'Red', 'NSW': 'Green', 'V': 'Red
', 'T': 'Red'}
```